

Learning Embodied Quadruped Agents for Posture-Aware Locomotion

Xiangyu Miao
Shanghai Jiao Tong
University
Shanghai, China
xiangyu.miao@outlook.com

Jun Sun*
Shanghai Aerospace
Control Technology
Institute
Shanghai, China
sjlovedh@hotmail.com

Hang Lai
Shanghai Jiao Tong
University
Shanghai, China
laihang99@sjtu.edu.cn

Xinpeng Di
Shanghai Aerospace
Control Technology
Institute
Shanghai, China
dixinpeng1@163.com

Jiahang Cao
Shanghai Jiao Tong
University
Shanghai, China
topsail@sjtu.edu.cn

Yong Yu
Shanghai Jiao Tong
University
Shanghai, China
yyu@cs.sjtu.edu.cn

Weinan Zhang*
Shanghai Jiao Tong
University
Shanghai, China
wnzhang@sjtu.edu.cn

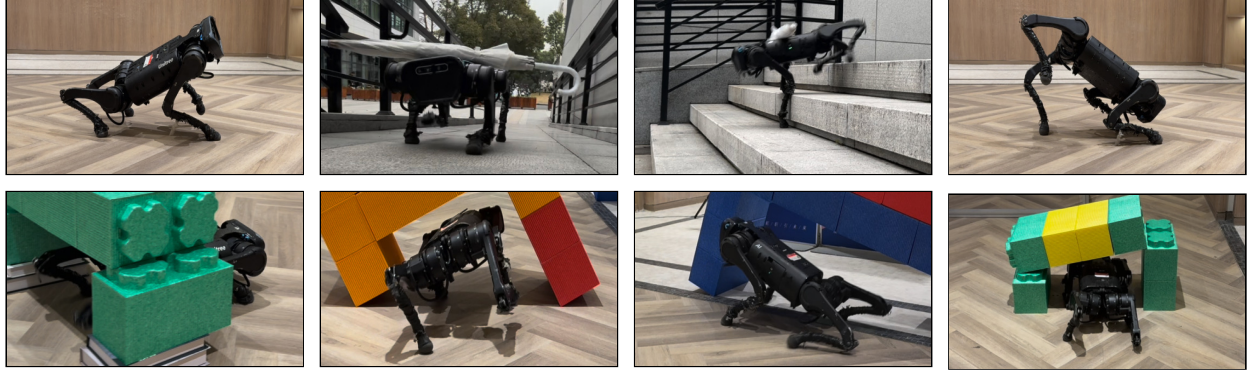


Figure 1: Deployment of the Posture-Aware Locomotion Agent (PALA) on Unitree A1 hardware across diverse real-world terrains. Highlighted tasks include traversal through stairs and slopes with neutral pitch, crawling through confined spaces with low clearance, and posture-modulated traversal through irregularly shaped cavities.

ABSTRACT

Recent advances in embodied intelligence have opened new directions for autonomous agents to operate in complex physical environments. In this work, we propose a Posture-Aware Locomotion Agent (PALA) trained via deep reinforcement learning. Unlike traditional quadruped agents that focus solely on velocity tracking, our agent learns to track task-oriented 6D motion commands, including linear and angular velocities, as well as desired body posture (height, pitch, and roll), in real time using only proprioceptive sensing and external commands. To improve robustness and terrain adaptability, we introduce two key heuristic designs: a progressive reward curriculum and an orientation command resampling

strategy. Combined with asymmetric actor-critic training, adversarial motion priors, and domain randomization, these components enable a single policy to generalize zero-shot across diverse and challenging environments. Additionally, we extend PALA to handle high-level instructions by integrating it with an autonomous agent powered by a large language model (LLM), enabling natural language task descriptions to be directly translated into executable 6D commands. Extensive simulation and real-world experiments demonstrate the responsiveness and accuracy of our posture-aware locomotion agent, underscoring its potential as a core component for embodied systems operating in unstructured settings.

CCS Concepts

• **Computer systems organization** → **Robotic control**; • **Computing methodologies** → **Reinforcement learning**; *Adversarial learning*; *Control methods*.

KEYWORDS

Embodied Intelligence, Reinforcement Learning, Quadruped Locomotion, Posture Control

*Co-corresponding authors.



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ACM ISBN 979-8-4007-2275-2/25/11
<https://doi.org/10.1145/3772429.3772430>

ACM Reference Format:

Xiangyu Miao, Jun Sun, Hang Lai, Xinpeng Di, Jiahang Cao, Yong Yu, and Weinan Zhang. 2025. Learning Embodied Quadruped Agents for Posture-Aware Locomotion. In *The Seventh International Conference on Distributed Artificial Intelligence (DAI '25), November 21–24, 2025, London, United Kingdom*. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3772429.3772430>

1 INTRODUCTION

Recent advances in artificial intelligence have propelled embodied agents to operate in complex physical environments [27]. In this context, quadruped robots serve as representative embodied agents that must interpret task directives and interact with complex terrains through physical motion. In quadruped robots, motion intelligence is typically decomposed into high-level planning (e.g., navigation) and low-level control (e.g., joint actuation). Designing effective locomotion policies remains a challenge due to high degrees of freedom (DoF) and the need to adapt across diverse terrains. Classical approaches rely on model-based control pipelines [14, 20, 29, 43], which require extensive system identification and expert knowledge for accurate modeling and tuning.

In contrast, deep reinforcement learning (RL) provides an end-to-end paradigm for agents to learn control policies by interacting with the environment, mapping observations directly to actions [13, 25, 33]. This enables adaptation without handcrafted dynamics and even handles faults like actuator degradation [39]. However, training directly on physical hardware is often impractical due to safety risks and low sample efficiency. Consequently, many adopt the *sim-to-real* paradigm: training in simulation, deploying on real hardware [17, 34, 46].

In standard quadrupedal RL setups, the agent is trained to track a 3-dimensional (3D) velocity command—linear velocities along the x - and y -axes and angular velocity around the z -axis in the local body frame. However, many real-world tasks require not only locomotion, but also explicit control over body posture. For instance, transporting objects requires pitch and roll regulation for stability. In terrain exploration, the agents may need to crouch or tilt their body to pass through narrow spaces or gain visibility. Despite these practical needs, most prior work focuses exclusively on velocity tracking [2, 11, 17, 24, 32], and some even penalize non-standard postures in favor of locomotion stability [12, 16, 26], thereby restricting the agent’s adaptability. For example, a policy optimized only for velocity tracking may fail to traverse confined spaces, adapt to irregular terrain geometries, and maintain operational visibility in cluttered environments. These limitations highlight the necessity of posture-aware locomotion for real-world applications.

To address this, we present **PALA**, a **P**osture-Aware Locomotion **A**gent trained via deep reinforcement learning. PALA enables quadruped agents to track high-dimensional motion commands—including *linear and angular velocities, body height, pitch, and roll*—using only proprioceptive feedback. The learned policy generalizes across diverse terrains and posture configurations, providing a foundation module for embodied agents in unstructured environments.

Our main contributions are as follows.

- (1) We introduce PALA, a posture-aware locomotion agent trained via deep reinforcement learning, capable of jointly

regulating velocity and body posture in a unified, end-to-end manner using only proprioceptive feedback.

- (2) We introduce two key heuristic designs—a progressive reward curriculum and an orientation command resampling strategy—that significantly improve tracking accuracy and terrain adaptability. Ablation studies confirm their critical contribution to posture-aware locomotion learning in complex environments.
- (3) We validate the zero-shot transferability of PALA from simulation to real-world quadruped platforms, and we demonstrate its potential as a general-purpose low-level control agent for embodied locomotion via an LLM-driven proof-of-concept, where natural language instructions are translated into executable commands without policy retraining.

2 RELATED WORK

2.1 Embodied Locomotion with Reinforcement Learning

Reinforcement learning (RL) has gained traction in quadruped locomotion, enabling agents to acquire complex motor behaviors via trial-and-error interaction. Prior work primarily focuses on tracking 3D velocity commands (linear in x/y , angular around z), enabling agents to reach target destinations [2, 7, 9, 11, 12, 16, 17, 24, 26, 32]. However, these policies often lack the flexibility to handle tasks requiring fine-grained posture control, such as crouching through narrow passages or adjusting pitch to gain visibility in cluttered or occluded environments.

To address these limitations, some approaches augment agent policies with gait-level control modules [3, 4, 22, 41], allowing transitions between predefined walking patterns such as trotting or crawling. While enhancing motion diversity, these methods fall short in enabling posture adaptation. Gait-level coordination lacks direct control over body orientation and height, which is essential for tasks involving confined environments, irregular structural constraints, and visibility-critical maneuvers.

This gap highlights the importance of posture-aware locomotion. While few studies explicitly focus on posture tracking, our work formulates it as an embodied agent problem: learning a low-level policy that simultaneously reasons over velocity and posture commands in a unified framework. In this view, posture tracking becomes a learned capability, not a handcrafted control heuristic.

2.2 Adversarial Motion Priors for Reward Shaping

Crafting effective RL rewards is challenging, especially for physical agents where desired behaviors are hard to specify. Rich reward formulations often require significant domain expertise and may lead to unintended behaviors if agents exploit poorly tuned objectives. To mitigate this, researchers have proposed regularization techniques that shape agent policies toward structured and energy-efficient movements [2, 12, 16, 23, 42].

A promising direction is the use of Adversarial Motion Priors (AMP), which leverage Generative Adversarial Imitation Learning (GAIL) [10] to implicitly encode behavior constraints. In this

framework, a discriminator network learns to differentiate agent-generated transitions from expert demonstrations. The adversarial signal acts as a "style reward", guiding agents toward plausible locomotion without hand-crafted priors or symmetry assumptions [6, 37, 38]. AMP thus transforms the motion prior problem into a learned reward shaping mechanism, helping embodied agents to acquire naturalistic motor patterns through implicit supervision.

In our approach, the reference motions used in AMP are generated via a *self-imitation pipeline*, where stable gaits learned without AMP are repurposed as expert demonstrations to guide subsequent adversarial training in more complex environments.

2.3 Asymmetric Learning with Privileged Inputs

Privileged learning [5] improves policy adaptability by leveraging simulation-only information during training. A common approach is the two-phase student-teacher paradigm [2, 15–18, 23, 24, 37], where an oracle teacher policy is trained with privileged inputs and then distilled into a deployable student policy. However, this scheme often suffers from low data efficiency, and the distilled student typically underperforms the teacher [26].

In contrast, we employ an *asymmetric actor-critic* framework [26, 28, 38, 44, 45] to address the limitations of the student-teacher paradigm. In our work, the actor receives only deployable observations (e.g., commands and proprioceptive feedback), while the critic has access to privileged simulation data during training. This enables robust behavior learning with privileged supervision—without the need for distillation.

3 METHODOLOGY

3.1 Reinforcement Learning Formulation

We formulate posture-aware locomotion as a sequential decision-making task under partial observability. Specifically, we model it as a Partially Observable Markov Decision Process (POMDP), defined by the tuple $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{P}, r, \gamma)$, where $\mathcal{S}$, $\mathcal{O}$, and $\mathcal{A}$ are the state, observation, and action spaces, respectively. $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ is the state transition probability, $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor.

An episode begins from an initial state s_0 . At each time step t , the agent receives a proprioceptive observation o_t , together with a 6D motion intent signal c_t . During training, c_t is randomly sampled to ensure broad coverage of the command space. During deployment, it may be issued by a human or generated by a high-level agent like an LLM (see Sec. 4.4). The low-level agent responds by taking action $a_t = \pi(o_t, c_t)$, influencing the environment's dynamics and receiving a scalar reward $r(s_t, a_t)$. The objective of RL is to find the optimal policy π_θ that maximizes the expected return:

$$\underset{\theta}{\text{maximize}} \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]. \quad (1)$$

3.1.1 Observation Space. Observation o_t includes angular velocity ω_t^z around the vertical axis, roll and pitch angles (ϕ_t, θ_t) obtained from on-device IMU, gravity projection g_t , DoF positions and velocities $(q_t, \dot{q}_t)$, and previous actions a_t . The command c_t consist

of target linear velocities $(v_{x,t}^{\text{cmd}}, v_{y,t}^{\text{cmd}})$, angular velocity $\omega_{z,t}^{\text{cmd}}$, body height h_t^{cmd} , pitch θ_t^{cmd} , and roll ϕ_t^{cmd} .

Orientation alignment is achieved by converting IMU-obtained quaternions $q = w + xi + yj + zk$ into Euler angles (ϕ, θ, ψ) using the following equations:

$$\begin{aligned} \phi &= \text{atan2}(2(wx + yz), 1 - 2(x^2 + y^2)), \\ \theta &= \text{asin}(2(wy - zx)), \\ \psi &= \text{atan2}(2(wz + xy), 1 - 2(y^2 + z^2)). \end{aligned} \quad (2)$$

Here, ϕ denotes roll, θ denotes pitch, and ψ denotes yaw.

3.1.2 Action Space. Action $a_t \in \mathbb{R}^{12}$ specifies joint position offsets relative to a predefined default configuration q^{default} . The resulting target joint positions are computed as $q_d = q^{\text{default}} + a_t$ and subsequently converted to motor torques via a Proportional-Derivative (PD) controller:

$$\tau = k_p(q_d - q) + k_d(\dot{q}_d - \dot{q}), \quad (3)$$

where q and $\dot{q}$ are the current joint positions and velocities, and q_d and $\dot{q}_d$ are the corresponding targets. In this setup, the target velocities $\dot{q}_d$ are set to zero, and the PD controller gains k_p and k_d determine the system's responsiveness and damping characteristics.

3.1.3 Reward Function. The reward at time t combines task tracking, style imitation, and regularization:

$$r_t = r_t^{\text{task}} + r_t^{\text{style}} + r_t^{\text{reg}}. \quad (4)$$

The task reward $r_t^{\text{task}} = \sum_{x \in \{v, \omega, h, \phi, \theta\}} R_x$ encourages compliance with 6D motion commands. The reward for linear velocity tracking is defined as:

$$R_v = \exp \left(- \frac{\|v_{x,y} - v_{x,y}^{\text{cmd}}\|^2}{\sigma_v} \right), \quad (5)$$

where σ_v is a scaling factor reflecting the reward's sensitivity to velocity error. The smaller σ_v , the more sensitive the reward is to the velocity error. Similarly, the reward for angular velocity tracking is as follows:

$$R_\omega = \exp \left(- \frac{\|\omega_z - \omega_z^{\text{cmd}}\|^2}{\sigma_\omega} \right). \quad (6)$$

Additional reward terms control body posture (i.e. height and orientation):

$$R_h = \exp \left(- \frac{|h - h^{\text{cmd}}|}{\sigma_h} \right), \quad (7)$$

$$R_x = \exp \left(- \frac{\|x - x^{\text{cmd}}\|^2}{\sigma_x} \right), \quad x \in \{\theta, \phi\}. \quad (8)$$

The style reward r_t^{style} comes from a discriminator D_ϕ trained via AMP [6, 37, 38] to distinguish agent transitions from reference demonstrations:

$$r_t^{\text{style}}(s_t, s_{t+1}) = \max[0, 1 - 0.25(D_\phi(s, s') - 1)^2], \quad (9)$$

where $D_\phi(s, s') \in [-1, 1]$ is the discriminator output classifying whether the transition (s_t, s_{t+1}) resembles reference behavior. The

reward encourages the agent to produce transitions that are indistinguishable from reference motions, thereby promoting stylistic fidelity during policy learning.

The discriminator is trained using a least-squares GAN loss with gradient penalty [6]:

$$L^{\text{Discriminator}}(\varphi) = \mathbb{E}_{(s,s') \sim \mathcal{D}} [(D_\varphi(s, s') - 1)^2] + \mathbb{E}_{(s,s') \sim \pi_\theta} [(D_\varphi(s, s') + 1)^2] + \frac{\omega_{gp}}{2} \mathbb{E}_{(s,s') \sim \mathcal{D}} [\|\nabla_\varphi D_\varphi(s, s')\|^2], \quad (10)$$

where $\mathcal{D}$ denotes the reference motion dataset and π_θ is the agent policy. The construction of reference data $\mathcal{D}$ used in AMP is detailed in the following section.

Regularization terms r_t^{reg} are similar to those used in [37], promoting energy efficiency, smooth joint dynamics, aerial phases, and collision avoidance.

3.2 Self-Imitated Motion Collection for Adversarial Training

We generate reference motions via a self-imitation pipeline. The agent is first trained without AMP on flat terrain using auxiliary rewards. Once stable gaits emerge—defined by tracking rewards exceeding a threshold (e.g., 90% of maximum)—we collect motion trajectories as expert data. These serve as style references in subsequent AMP-based training, where the agent learns to generalize posture-aware locomotion across varied terrains using adversarial guidance. This design leverages the relative ease of achieving stable locomotion on flat terrain via regularization, which becomes non-trivial on unstructured surfaces. Moreover, collecting motion data directly from the agent in simulation avoids the embodiment gap and eliminates the need for motion retargeting—a common challenge in conventional AMP pipelines.

3.3 Asymmetric Actor-Critic Learning

PALA uses a history encoder to process a 5-step observation-action sequence, producing a latent representation and an estimate of the base linear velocity. The choice of history length is further investigated in Sec. 4.3, where we analyze its impact on posture-aware locomotion performance. The resulting latent vector l_t and velocity estimate are concatenated with the 6D command c_t and passed to the actor, which produces 12D joint targets. These are subsequently converted to motor torques by a PD controller. Meanwhile, the critic receives privileged information o_t^{priv} , including base linear velocities (v_x, v_y), base angular velocity ω_z , body pitch and roll (θ, ϕ), gravity projection, 6D command, DoF positions and velocities, previous actions, local heightmap oriented along the agent's facing direction (sampled over a 17×11 grid spanning $x \in [-0.8, 0.8]$ and $y \in [-0.5, 0.5]$, covering a $1 \text{ m} \times 1.6 \text{ m}$ area around the agent), friction, restitution coefficient, base mass, center of mass (CoM) position, contact forces, and Boolean indicators of undesired link collisions.

We optimize the actor and the critic using Proximal Policy Optimization (PPO) [31]. In addition, we include a supervised loss for the linear velocity estimator produced by the history encoder. The

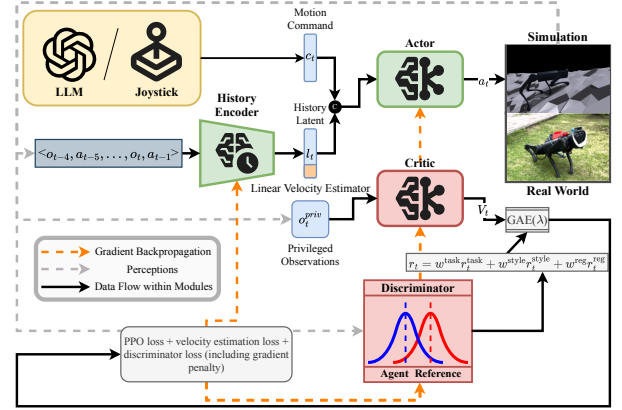


Figure 2: Overview of the PALA framework. The actor, encoder, critic, and discriminator are implemented as Multi-Layer Perceptrons (MLPs). Module colors indicate deployment scope: **Green modules** are deployable on physical hardware, **red modules** are used exclusively for simulator-based training, and **yellow module** represents the human-agent interaction interface (via LLM/joystick).

ground-truth base linear velocity $v_{\text{base}}^{\text{gt}}$ is available from the simulation environment, enabling direct supervision via mean squared error:

$$L^{\text{Est}} = \frac{1}{2} \mathbb{E} [\|\hat{v}_{\text{base}} - v_{\text{base}}^{\text{gt}}\|^2], \quad (11)$$

where $\hat{v}_{\text{base}}$ denotes the estimated base linear velocity, and $v_{\text{base}}^{\text{gt}}$ is the corresponding ground-truth value. This auxiliary objective improves the policy's awareness of its own kinematic state.

3.4 Training Curricula

To stabilize early RL, we adopt curriculum learning [19] to gradually increase task difficulty based on policy performance. Our approach includes terrain, reward, and command curricula.

Terrain Curriculum. We simulate diverse terrains—wavy surfaces, rough slopes, ascending and descending stairs, discrete obstacles, and flat rough terrain [30]—to reflect urban and wilderness settings. A shared difficulty factor adjusts terrain parameters such as wave amplitude, slope steepness, stair height, and obstacle size. Each terrain type spans 10 difficulty levels (0–1). Agents begin on the easiest level and progress when they traverse more than half the terrain length. Demotion occurs only if displacement falls significantly below the expected value (based on commanded velocity and episode duration). This adaptive scheme promotes terrain-aware locomotion while maintaining training stability.

Reward Curriculum. We use a staged reward curriculum for posture control. Early training focuses solely on locomotion, freezing posture rewards. Midway, posture tracking rewards are gradually introduced, allowing agents to learn body adjustment alongside movement. In later stages, the full reward function integrates both

Table 1: Randomized Variables and Sampling Ranges

Variable	Minimum	Maximum	Unit
Ground friction	0.05	2.75	-
Restitution	0	1.0	-
Load mass	0	3.0	kg
Link mass	80	120	%
CoM position	-0.05	0.05	m
PD controller P-gain	80	120	%
PD controller D-gain	80	120	%
Motor power	80	120	%
Action delay	0.00	0.02	s

locomotion and posture objectives. This progression avoids instability from early multitask control on complex terrains.

Command Curriculum. During training, 6D commands are randomly sampled. For most terrains, pitch and roll follow $\mathcal{N}(0, 0.25)$, and body height is uniformly sampled from 0.1–0.4 m. To account for physical constraints of the agent hardware, pitch angles are constrained to the range $[-\frac{\pi}{4}, \frac{\pi}{4}]$ radians, and roll angles are constrained to $[-\frac{\pi}{6}, \frac{\pi}{6}]$ radians. For stairs:

- **Ascending stairs:** Pitch commands follow a normal distribution with a mean of -0.5 radians and a standard deviation of 0.25 radians, further clipped to $[-\frac{\pi}{4}, 0]$ radians.
- **Descending stairs:** Pitch commands follow a normal distribution with a mean of 0.5 radians and a standard deviation of 0.25 radians, further clipped to $[0, \frac{\pi}{4}]$ radians.

Posture-aware locomotion under these constraints is challenging. Following [23], we dynamically expand the velocity bounds and reduce disturbance intervals when the velocity tracking reward exceeds 80% of its maximum. Additionally, we initialize a larger sampling range for agents on flat terrain to encourage agile, dynamic behavior.

3.5 Domain Randomization

To bridge the sim-to-real gap, we apply domain randomization [6, 8, 12, 15, 16, 22, 23, 34, 35, 37, 40, 42, 46, 47], varying training conditions to reflect real-world distributions. Parameters such as ground friction, restitution, body mass, action delay, PD gains, and motor torque are sampled from uniform ranges. We also inject random velocity disturbances to simulate external perturbations and improve robustness. Sampling ranges are detailed in Table 1.

4 EXPERIMENTAL RESULTS

We evaluate the effectiveness of our proposed agent PALA through extensive simulation and real-world experiments. Both training and evaluation were conducted in IsaacGym [21], leveraging its parallel simulation capabilities. We train 4096 agents in parallel across heterogeneous terrain distributions on an NVIDIA RTX 3080 Ti GPU. The physics simulation step size was set to $\Delta t = 0.005$ s (200 Hz), with an action decimation factor of 4, resulting in an effective control frequency of 50 Hz. Early termination was applied upon collision event on robot base. The evaluation is structured around the following key questions:

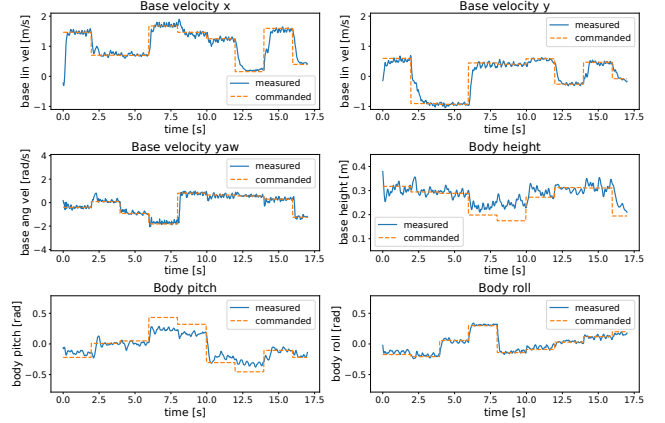


Figure 3: 6D command tracking performance on flat simulated terrain. The policy demonstrates accurate command tracking while maintaining high-level responsiveness.

- **Q1:** Can an embodied PALA reliably track motion commands for both locomotion and posture modulation?
- **Q2:** What architectural and algorithmic components are critical to the robustness and adaptability of the learned policy?
- **Q3:** Can PALA serve as a general-purpose low-level control module for high-level agents, such as those driven by natural language instructions via large language models?
- **Q4:** How well does the policy generalize from simulation to real-world deployment across diverse and unstructured terrains?

We first address **Q1** by evaluating how well the agent tracks high-dimensional control commands through two lenses: general responsiveness under dynamic intent changes, and posture precision across a range of orientation command combinations.

4.1 Policy Responsiveness

To evaluate the agent’s tracking capacity under dynamic intents, we deployed it on flat terrain where the 6D command vector—comprising $(v_x, v_y, \omega_z, h, \theta, \phi)$ —was resampled every 2 seconds. As shown in Figure 3, the policy consistently achieved accurate tracking of both translational and rotational commands, including rapid posture reconfigurations and transitions between running, crawling, crouching, and in-place turning.

We observed high-fidelity tracking across all control dimensions, with minimal delay or overshoot. A notable deviation occurred between 6–10 seconds, where the agent was commanded to maintain a low-height, positive-pitch posture while executing high-speed locomotion. This combination presents inherent physical challenges due to the robot’s morphology—sustaining positive pitch and low height during rapid movement is mechanically constrained. Despite this, the agent preserved gait stability and command responsiveness, demonstrating the adaptability of the learned policy under physically demanding conditions.

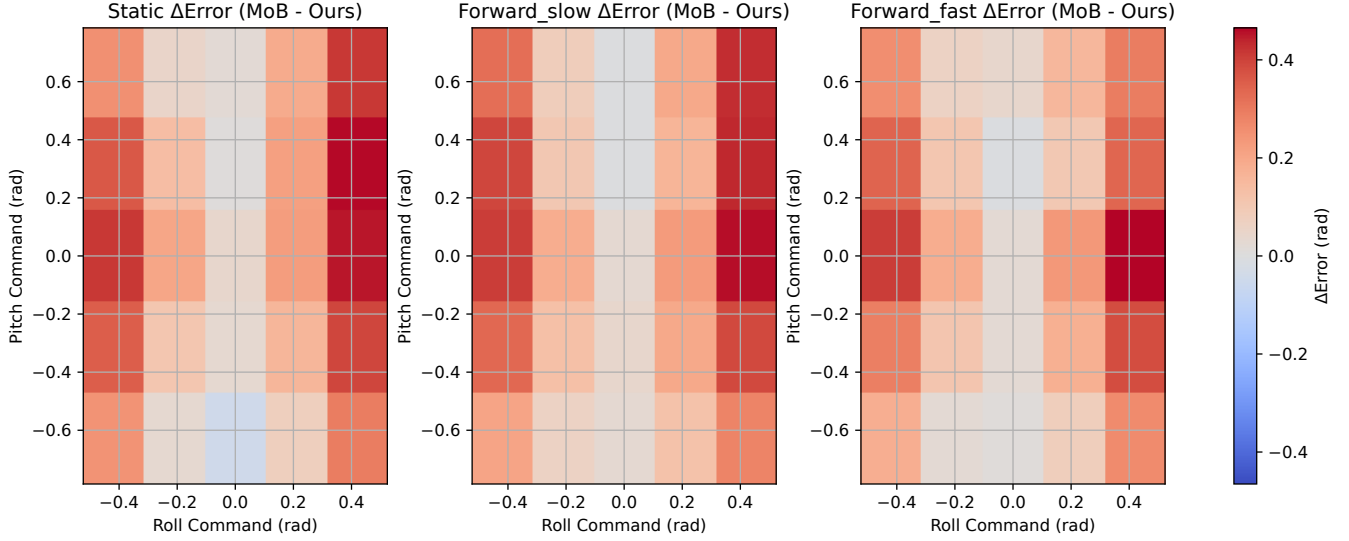


Figure 4: Tracking error difference between PALA and MoB across various orientation commands and forward velocities. Positive values indicate superior accuracy by PALA, whereas negative values indicate the opposite.

4.2 Orientation Tracking

To further evaluate PALA’s capability in posture control, we conduct a comparative study against MoB [22], a framework that enables user control over gait and body orientation. Unlike PALA, MoB employs a different reward formulation for orientation tracking and is trained exclusively on flat terrain, limiting its generalization to more complex environments. For a fair comparison, we retrained a PALA policy on the same embodied platform as MoB. We evaluated both policies in simulation on flat terrain using identical 6D command configurations. Specifically, we tested the tracking performance of various (θ, ϕ) command pairs under three forward velocity settings: static (0 m/s), slow (0.5 m/s), and fast (1.0 m/s). To ensure statistical reliability, each command combination was evaluated over five independent trials, and the results were averaged to compute the final tracking error metrics. To quantify and compare the orientation tracking accuracy of both methods, we compute a combined error metric by taking the root mean square of the pitch and roll RMSEs:

$$\text{Orientation Tracking Error} = \sqrt{\frac{1}{2} (\text{RMSE}_\theta^2 + \text{RMSE}_\phi^2)},$$

where each RMSE term is computed as:

$$\text{RMSE}_x = \sqrt{\frac{1}{T} \sum_{t=1}^T (x_t^{\text{cmd}} - x_t)^2}, \quad x \in \{\theta, \phi\},$$

with T denoting the number of evaluation steps per episode.

The results are visualized in Figure 4, where the horizontal and vertical axes denote roll and pitch commands, respectively. Color intensity encodes the performance difference: red regions indicate combinations where PALA achieved lower error (i.e., better tracking accuracy), while blue regions indicate the opposite.

From the heatmaps, we observe that PALA consistently outperforms MoB across most orientation command combinations and velocity conditions. This advantage is particularly pronounced in roll tracking and at extreme posture angles near the boundaries of the command space. Notably, MoB struggles to track roll commands with high precision, likely due to the coupling between pitch and roll control in its architecture. In contrast, our decoupled control strategy facilitates more accurate and independent posture regulation. While MoB occasionally achieves slightly better tracking in certain configurations—such as $(\theta = -\frac{\pi}{4}, \phi = 0)$ under static conditions—the overall performance gap remains marginal.

Overall, this comparison underscores the superior generalization and control precision enabled by our method, especially in demanding posture configurations and under dynamic locomotion conditions.

4.3 Ablation Study

To address Q2, we perform controlled simulation experiments to isolate the contribution of each design component. Evaluation is based on *mean per-step reward*, averaged over five trials unless otherwise noted. Velocity tracking rewards are scaled to a maximum of 1.0, while posture-related rewards are capped at 0.5 to prioritize locomotion fidelity.

Unless stated otherwise, evaluation follows two distinct protocols depending on the terrain type:

- **Flat terrain:** We adopt the same setup as in Sec. 4.1, where all dimensions of the 6D command vector are randomly resampled every 2 s. Each policy variant is evaluated over five trials with different random seeds, and the results are averaged.
- **Non-flat terrains:** Velocity commands are fixed, while posture commands are randomly resampled every 2 s. Testing

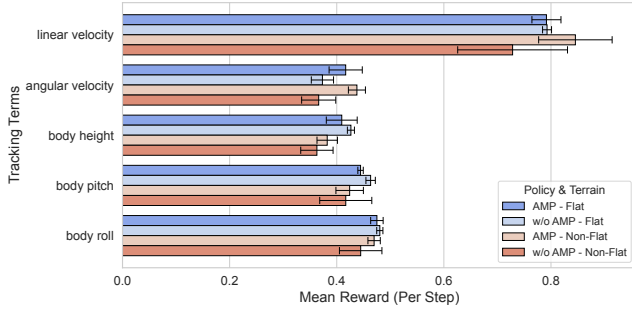


Figure 5: Comparison of policies with and without AMP across flat and non-flat terrains. AMP significantly improves velocity tracking and overall locomotion quality in complex environments.

is performed once on each of the five challenging terrains—wavy, slope, stairs up, stairs down, and discrete obstacles—and the aggregated results are used to compute the mean and variance. Error bars in figures represent the standard deviation across trials.

Our ablation study comprises three parts: (i) assessing AMP’s impact on policy quality, (ii) comparing **PALA** with a **naïve RL baseline**, where the agent tracks 6D commands without the proposed heuristics, and (iii) analyzing history encoder variants with different temporal horizons.

Adversarial Motion Priors: Following the evaluation protocol described above, we compare two policy variants—one trained with AMP and one without—across flat and non-flat terrains. As shown in Figure 5, both policies perform similarly on flat terrain. However, in more challenging environments involving stairs, obstacles, and uneven surfaces, the AMP-enhanced policy achieves significantly better tracking performance, particularly in velocity tracking. This improvement stems from AMP’s ability to regularize the agent’s movements, resulting in smoother, more coherent gait patterns. In contrast, the non-AMP policy exhibits more unstable or erratic movements, leading to degraded tracking fidelity and premature episode termination. These results highlight AMP’s critical contribution to robust and natural locomotion under real-world terrain variability.

Reward Curriculum and Orientation Command Strategy: We compare the proposed **PALA** framework against a variant (**PALA-variant**) trained with the same RL recipe but without two of our key heuristic designs: (1) the progressive reward curriculum, and (2) the complex-terrain orientation command resampling strategy. Under the unified evaluation protocol, we select one of the most challenging posture-aware locomotion tasks: stair traversal while tracking a fixed velocity command and resampling posture commands every 2 s. As shown in Figure 6, PALA consistently outperforms its variant across all tracking dimensions on this task. Without our heuristics, the PALA-variant struggles to maintain stable gaits and accurate posture regulation, leading to larger tracking errors and occasional failure to complete the

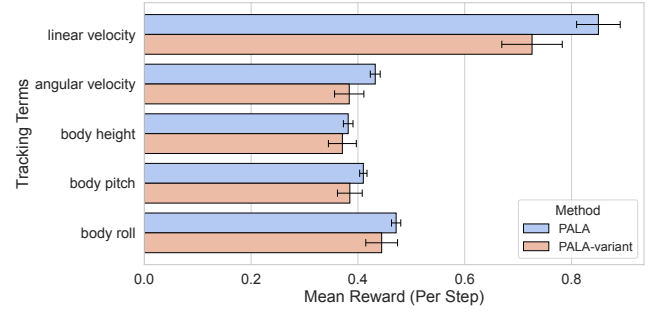


Figure 6: Comparison between PALA and PALA-variant in stair traversal under dynamic posture modulation. The PALA framework achieves consistently higher tracking rewards across all control dimensions.

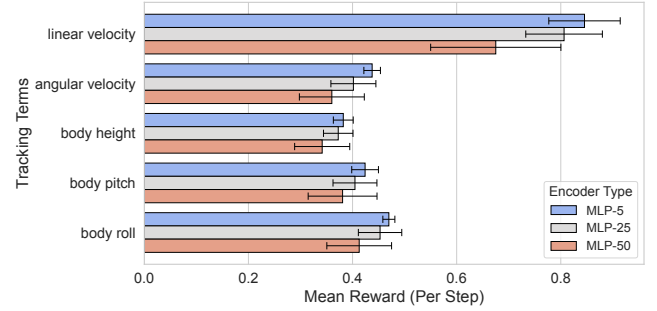


Figure 7: Comparison of encoder variants with different temporal horizons. MLP-5 achieves the highest average tracking rewards across all control dimensions, indicating that short-term history is most effective for posture-aware locomotion.

terrain. These results highlight the importance of our curriculum and command sampling strategy in enhancing adaptability and robustness on complex terrains.

History Length Analysis: To investigate the impact of history length on posture-aware locomotion, we compare three MLP-based encoder configurations under the unified evaluation protocol: **MLP-5** (5-step observation-action history), **MLP-25**, and **MLP-50**. All models share the same latent dimension, isolating the impact of history length from model capacity. We evaluate the policies across five challenging non-flat terrains. The results in Figure 7 show that MLP-5 achieves the best performance across all tracking terms. This indicates that the key information required for posture-aware locomotion is compactly contained within short-term history, and that incorporating excessively long histories may occupy the policy’s memory capacity and degrade performance.

4.4 LLM-Driven Control (Proof-of-Concept)

To further demonstrate the plug-and-play capability of PALA as a generic low-level control module (Q3), we conduct a proof-of-concept experiment where high-level task specifications are provided in natural language and translated into executable 6D motion

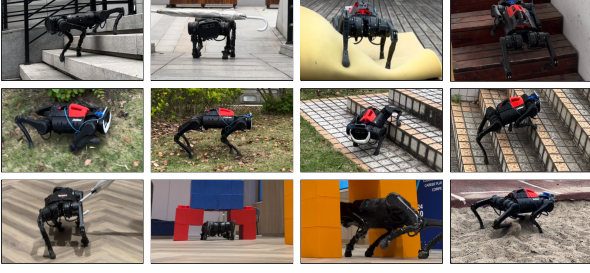


Figure 8: Real-world deployment snapshots of PALA, demonstrating its ability to simultaneously control locomotion and posture across diverse terrains.

commands by a large language model (LLM). In this setting, a GPT-4o-mini model [1] is prompted with task descriptions (e.g., “Crawl with medium speed and low body height”), and outputs structured locomotion directives comprising target linear and angular velocities, body height, pitch, and roll. These commands are then fed directly to PALA without any policy retraining. An example GPT-4o-mini prompt used for this translation is provided in Appendix A.

Unlike a fixed-rate command interface, the LLM–PALA connection is event-driven: high-level commands are updated only when the user issues a new task or modifies the ongoing one. Once a command is received, PALA continuously executes it at 50 Hz, ensuring smooth and responsive low-level control until the next high-level instruction is provided. We employ a few-shot prompting strategy, where the LLM is given examples mapping textual instructions to corresponding 6D configurations, along with constraints on feasible ranges for each control dimension. This enables the model to infer suitable parameters for novel instructions while respecting the physical limits of the robot.

We evaluate this setup in IsaacGym [21] by issuing multi-step natural language commands covering various locomotion and posture scenarios, such as running, turning, crawling, stair climbing with posture stabilization, and combined pitch-roll adjustment tasks. Across trials, PALA successfully executed LLM-generated commands with high fidelity, validating its ability to serve as a drop-in low-level controller for diverse high-level agents, including those based on natural language reasoning. A visual example of this interaction is provided in Appendix B.

4.5 Real-World Deployment

To address Q4, we deployed the learned low-level policy on a Unitree A1 quadruped robot [36] zero-shot. The control frequency was set to 50 Hz with PD gains configured as $k_p = 28$ and $k_d = 0.7$.

We tested the embodied agent across diverse physical terrains, including grass, rubber mats, sponge pads, slopes, sand, wooden stairs, tiled steps, and confined spaces. Figure 1 and Figure 8 show representative snapshots. Beyond standard terrain traversal, we designed several challenging scenarios to evaluate posture-aware locomotion control in the real world. These include asymmetric cavities requiring targeted roll and pitch configurations, and low-clearance passages that demand crawling behavior. In slope and stair traversal tasks, we loosely placed an external object on the

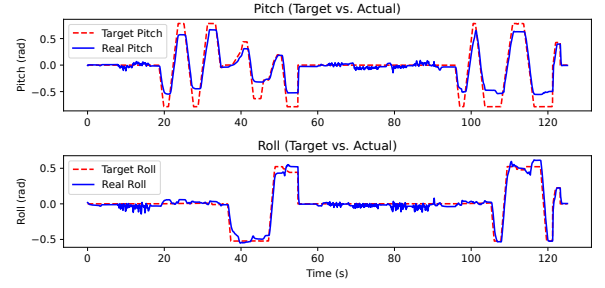


Figure 9: Real-time tracking of pitch and roll angles in the physical robot. The policy provides reliable posture modulation under real-world conditions.

robot’s back without additional fixation, demonstrating its ability to maintain neutral pitch under perturbation. We also tested the agent’s resilience to external disturbances and deformable surfaces, such as sponge pads and sand. Across all trials, the agent successfully completed the tasks with a 100% success rate, highlighting its robustness and adaptability in unstructured environments. These results confirm the real-world viability of PALA.

To further assess posture tracking responsiveness and accuracy, we recorded IMU-derived pitch and roll angles. As shown in Figure 9, the agent achieved precise alignment with target body orientations in real time, even during dynamic transitions. Despite differences in dynamics and sensor noise, the agent demonstrates its robustness under domain shift.

5 CONCLUSION

In this work, we present PALA, a Posture-Aware Locomotion Agent trained via deep reinforcement learning. Our approach focuses on learning a robust and generalizable low-level policy that enables simultaneous locomotion and posture control across diverse and challenging terrains, without relying on visual or semantic input. To enhance policy responsiveness and terrain adaptability, we integrate AMP into an asymmetric actor-critic framework and introduce two key heuristic designs: a progressive reward curriculum and an orientation command resampling strategy, validated through targeted ablation studies.

Beyond its standalone performance in posture-aware locomotion, our LLM-driven proof-of-concept experiment highlights PALA’s versatility as a modular low-level controller that can seamlessly integrate with high-level agents. By bridging natural language instructions to fine-grained motor control, this integration exemplifies a hierarchical embodied AI architecture in which semantic reasoning and precise locomotion are handled by specialized, distributed components. Such modularity aligns closely with the principles of distributed artificial intelligence, and opens avenues for deploying PALA in broader multi-agent and multi-modal decision-making systems.

ACKNOWLEDGEMENTS

This work was supported in part by the National Natural Science Foundation of China under Grant No. U22B2042.

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A PROMPT EXAMPLE

Below is the prompt used to instruct the GPT-4o-mini model for translating natural language tasks into 6D control commands for PALA:

You are a high-level agent that generates control commands for a quadruped robot based on a task description.

The command is a 6-dimensional vector including:

1. Linear velocity along x-axis (forward/backward) [m/s]
2. Linear velocity along y-axis (lateral) [m/s]
3. Angular velocity around z-axis (yaw rate) [rad/s]
4. Height offset from default 0.25 m [m]
5. Pitch angle [rad], range $[-0.5, 0.5]$
6. Roll angle [rad], range $[-0.5, 0.5]$

The pitch controls forward/backward tilt, roll controls left/right tilt. The output should be a Python list in the format: `[x_vel, y_vel, yaw_vel, height, pitch, roll]`

Example:

Task: "Move forward quickly and lower the body slightly"

Output: `[0.8, 0.0, 0.0, -0.05, 0.0, 0.0]`

Task: "Return to normal state"

Output: `[0.0, 0.0, 0.0, 0.0, 0.0, 0.0]`

Now generate the command for the following task:

Task: "{task_description}"

Output:

Remarks. The placeholder {task_description} is replaced at runtime by the user-specified natural language instruction (e.g., "Crawl with low body height" or "Move forward with negative pitch"). If occasional ambiguity arises in the LLM-generated commands, for example due to vague or underspecified task descriptions, it can be mitigated through prompt refinement, adding explicit range constraints in the examples, or introducing a confirmation stage before sending commands to PALA.

B LLM-CONTROLLED MOTION SEQUENCE VISUALIZATION

To visualize the behavior of PALA under LLM-generated commands, we present a motion sequence driven by the following multi-step instruction:

- (1) Stay static.
- (2) Walk forward slowly.
- (3) Move forward and lower your body to crouch.
- (4) Stop.
- (5) Lean forward.
- (6) Lean backward.
- (7) Return to default state and run quickly forward.

Figure 10 overlays keyframes extracted from the simulation.

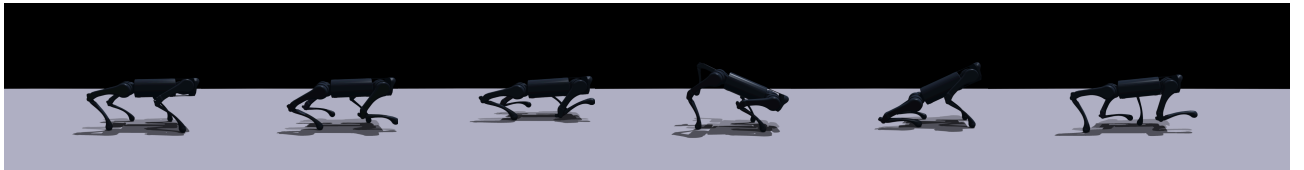


Figure 10: Motion sequence executed by PALA in response to structured 6D control commands generated by an LLM. The agent begins from a static pose, walks slowly, crouches, pauses, modulates pitch forward and backward, returns to default posture, and accelerates forward.

C REWARD FUNCTION

Table 2 provides a detailed overview of the reward structure utilized in our work.

Term	Expression	Weight	Tracking σ
Linear velocity tracking	$\exp\left(-\frac{\ v_{x,y}-v_{x,y}^{\text{cmd}}\ ^2}{\sigma_v}\right)$	1.0	0.15
Angular velocity tracking	$\exp\left(-\frac{\ \omega_z-\omega_z^{\text{cmd}}\ ^2}{\sigma_\omega}\right)$	0.5	0.15
Body height tracking	$\exp\left(-\frac{ h-h^{\text{cmd}} }{\sigma_h}\right)$	0.5	0.15
Body pitch tracking	$\exp\left(-\frac{\ \theta-\theta^{\text{cmd}}\ ^2}{\sigma_\theta}\right)$	0.5	0.1
Body roll tracking	$\exp\left(-\frac{\ \phi-\phi^{\text{cmd}}\ ^2}{\sigma_\phi}\right)$	0.5	0.05
Torque	$\ \tau\ ^2$	-0.0001	-
DoF acceleration	$\ \ddot{q}\ ^2$	$-2.5e^{-7}$	-
Feet air time	$T_{\text{air}} - 0.5$	0.5	-
Action rate	$\ a_t - a_{t-1}\ ^2$	-0.01	-
AMP	Sec 3.1	0.01	-

Table 2: Details of the reward terms used in our training process.